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1. Create a directory named `workspace` in your home folder.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~$ mkdir workspace
digvijay@digvijay-virtualbox:~$ ls
Assignment  bda  Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos  workspace
digvijay@digvijay-virtualbox:~$ cd workspace/
digvijay@digvijay-virtualbox:~/workspace$
```

2. Inside `workspace`, create two subdirectories named `source` and `backup`.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ mkdir source backup
digvijay@digvijay-virtualbox:~/workspace$ ls
backup  source
digvijay@digvijay-virtualbox:~/workspace$
```

3. Create three empty files in `source` named `alpha.txt`, `beta.txt`, and `gamma.txt`.

```
digvijay@digvijay-virtualbox: ~/workspace/source
digvijay@digvijay-virtualbox:~/workspace/source$ touch alpha.txt beta.txt gamma.txt
digvijay@digvijay-virtualbox:~/workspace/source$ ls
alpha.txt  beta.txt  gamma.txt
digvijay@digvijay-virtualbox:~/workspace/source$
```

4. Rename `alpha.txt` to `delta.txt`.

```
digvijay@digvijay-virtualbox: ~/workspace/source
digvijay@digvijay-virtualbox:~/workspace/source$ ls
alpha.txt  beta.txt  gamma.txt
digvijay@digvijay-virtualbox:~/workspace/source$ mv alpha.txt delta.txt
digvijay@digvijay-virtualbox:~/workspace/source$ ls
beta.txt  delta.txt  gamma.txt
digvijay@digvijay-virtualbox:~/workspace/source$
```

5. Move `delta.txt` from the `source` directory to the `backup` directory.

```
digvijay@digvijay-virtualbox: ~/workspace/backup
digvijay@digvijay-virtualbox:~/workspace/source$ ls
beta.txt  delta.txt  gamma.txt
digvijay@digvijay-virtualbox:~/workspace/source$ mv delta.txt ~/workspace/backup/
digvijay@digvijay-virtualbox:~/workspace/source$ cd ..
digvijay@digvijay-virtualbox:~/workspace$ cd backup/
digvijay@digvijay-virtualbox:~/workspace/backup$ ls
delta.txt
digvijay@digvijay-virtualbox:~/workspace/backup$
```

6. Create a new user named `intern`.

```
digvijay@digvijay-virtualbox: ~/workspace/backup
digvijay@digvijay-virtualbox:~/workspace/backup$ sudo adduser intern
[sudo] password for digvijay:
info: Adding user `intern' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `intern' (1001) ...
info: Adding new user `intern' (1001) with group `intern (1001)' ...
info: Creating home directory `/home/intern' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for intern
Enter the new value, or press ENTER for the default
        Full Name []: intern
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `intern' to supplemental / extra groups `users' ...
info: Adding user `intern' to group `users' ...
digvijay@digvijay-virtualbox:~/workspace/backup$
```

7. Create a new group named `training`.

```
digvijay@digvijay-virtualbox: ~/workspace/backup
digvijay@digvijay-virtualbox:~/workspace/backup$ sudo groupadd training
digvijay@digvijay-virtualbox:~/workspace/backup$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,digvijay
```

```
vboxsf:x:123:
intern:x:1001:
training:x:1002:
digvijay@digvijay-virtualbox:~/workspace/backup$
```

8. Add the user `intern` to the `training` group.

```
digvijay@digvijay-virtualbox: ~/workspace/backup
digvijay@digvijay-virtualbox:~/workspace/backup$ sudo usermod -aG training intern
digvijay@digvijay-virtualbox:~/workspace/backup$ groups intern
intern : intern users training
digvijay@digvijay-virtualbox:~/workspace/backup$
```

9. Change the password for the user `intern`.

```
digvijay@digvijay-virtualbox: ~/workspace/backup
digvijay@digvijay-virtualbox:~/workspace/backup$ sudo passwd intern
New password:
Retype new password:
passwd: password updated successfully
digvijay@digvijay-virtualbox:~/workspace/backup$
```

10. Create a file named `private.txt` and set permissions so only the owner can read and write to it ($rw-------$).

```
digvijay@digvijay-virtualbox: ~/workspace/source
digvijay@digvijay-virtualbox:~/workspace/source$ touch private.txt
digvijay@digvijay-virtualbox:~/workspace/source$ ls -l private.txt
-rw-rw-r-- 1 digvijay digvijay 0 Feb 27 22:52 private.txt
digvijay@digvijay-virtualbox:~/workspace/source$ sudo chmod 600 private.txt
digvijay@digvijay-virtualbox:~/workspace/source$ ls -l private.txt
-rw------- 1 digvijay digvijay 0 Feb 27 22:52 private.txt
digvijay@digvijay-virtualbox:~/workspace/source$
```

11. Create a file named `public.txt` and set permissions so everyone can read it, but only the owner can write to it ($rw-r--r--$).

```
digvijay@digvijay-virtualbox: ~/workspace/source
digvijay@digvijay-virtualbox:~/workspace/source$ touch public.txt
digvijay@digvijay-virtualbox:~/workspace/source$ ls -l public.txt
-rw-rw-r-- 1 digvijay digvijay 0 Feb 27 22:56 public.txt
digvijay@digvijay-virtualbox:~/workspace/source$ sudo chmod 644 public.txt
digvijay@digvijay-virtualbox:~/workspace/source$ ls -l public.txt
-rw-r--r-- 1 digvijay digvijay 0 Feb 27 22:56 public.txt
digvijay@digvijay-virtualbox:~/workspace/source$
```

12. Create a file named `script.sh` and give it execute permissions for the owner, group, and others ($rwxr-xr-x$).

```
digvijay@digvijay-virtualbox: ~/workspace/source
digvijay@digvijay-virtualbox:~/workspace/source$ touch script.sh
digvijay@digvijay-virtualbox:~/workspace/source$ ls -l script.sh
-rw-rw-r-- 1 digvijay digvijay 0 Feb 27 22:57 script.sh
digvijay@digvijay-virtualbox:~/workspace/source$ sudo chmod 755 script.sh
digvijay@digvijay-virtualbox:~/workspace/source$ ls -l script.sh
-rwxr-xr-x 1 digvijay digvijay 0 Feb 27 22:57 script.sh
digvijay@digvijay-virtualbox:~/workspace/source$
```

13. Change the owner of `backup/delta.txt` to the user `intern`.

```
digvijay@digvijay-virtualbox: ~/workspace/backup
digvijay@digvijay-virtualbox:~/workspace/backup$ ls -l delta.txt
-rw-rw-r-- 1 digvijay digvijay 0 Feb 27 22:36 delta.txt
digvijay@digvijay-virtualbox:~/workspace/backup$ sudo chown intern delta.txt
digvijay@digvijay-virtualbox:~/workspace/backup$ ls -l delta.txt
-rw-rw-r-- 1 intern digvijay 0 Feb 27 22:36 delta.txt
digvijay@digvijay-virtualbox:~/workspace/backup$
```

14. Change the group ownership of the `workspace` directory to `training`.

```
digvijay@digvijay-virtualbox: ~
digvijay@digvijay-virtualbox:~$ ls -l workspace/
total 8
drwxrwxr-x 2 digvijay digvijay 4096 Feb 27 22:42 backup
drwxrwxr-x 2 digvijay digvijay 4096 Feb 27 22:57 source
digvijay@digvijay-virtualbox:~$ sudo chgrp -R training workspace/
digvijay@digvijay-virtualbox:~$ ls -l workspace/
total 8
drwxrwxr-x 2 digvijay training 4096 Feb 27 22:42 backup
drwxrwxr-x 2 digvijay training 4096 Feb 27 22:57 source
digvijay@digvijay-virtualbox:~$
```

15. Delete the file `source/beta.txt`.

```
digvijay@digvijay-virtualbox: ~/workspace/source
digvijay@digvijay-virtualbox:~/workspace/source$ ls
beta.txt  gamma.txt  private.txt  public.txt  script.sh
digvijay@digvijay-virtualbox:~/workspace/source$ rm beta.txt
digvijay@digvijay-virtualbox:~/workspace/source$ ls
gamma.txt  private.txt  public.txt  script.sh
digvijay@digvijay-virtualbox:~/workspace/source$
```

16. Delete the `source` directory and all files remaining inside it using a single command.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ ls
backup  source
digvijay@digvijay-virtualbox:~/workspace$ rm -r source/
digvijay@digvijay-virtualbox:~/workspace$ ls
backup
digvijay@digvijay-virtualbox:~/workspace$
```

17. List all files in the `/etc` directory and save the output to a file named `etc_list.txt`.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:/$ ls -l etc > ~/workspace/etc_list.txt
digvijay@digvijay-virtualbox:/$ cd ~/workspace/
digvijay@digvijay-virtualbox:~/workspace$ ls
backup  etc_list.txt
digvijay@digvijay-virtualbox:~/workspace$ cat etc_list.txt
total 1128
-rw-r--r--  1 root root     3444 Jul  5  2023 adduser.conf
-rw-r--r--  1 root root       44 Feb 26 23:22 adjtime
drwxr-xr-x  3 root root     4096 Aug  5  2025 alsa
drwxr-xr-x  2 root root     4096 Feb 26 23:24 alternatives
-rw-r--r--  1 root root      335 Apr  8  2024 anacrontab
drwxr-xr-x  5 root root     4096 Aug  5  2025 apm
drwxr-xr-x  2 root root     4096 Aug  5  2025 apparmor
drwxr-xr-x  9 root root     4096 Aug  5  2025 apparmor.d
drwxr-xr-x  3 root root     4096 Aug  5  2025 apport
drwxr-xr-x  8 root root     4096 Feb 26 23:23 apt
drwxr-xr-x  3 root root     4096 Aug  5  2025 avahi
-rw-r--r--  1 root root     2319 Mar 31  2024 bash.bashrc
-rw-r--r--  1 root root       45 Jan 25  2020 bash_completion
```

18. Use a command to see the last 10 lines of the `/etc/passwd` file.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ tail /etc/passwd
saned:x:113:116::/var/lib/saned:/usr/sbin/nologin
geoclue:x:114:117::/var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:115:113::/nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
polkitd:x:989:989:User for polkitd:/:/usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
colord:x:118:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
nm-openvpn:x:119:121:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
digvijay:x:1000:1000:Digvijay:/home/digvijay:/bin/bash
intern:x:1001:1001:intern,,,:/home/intern:/bin/bash
digvijay@digvijay-virtualbox:~/workspace$
```

19. Use a command to see the first 5 lines of the `/etc/group` file.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ head -n 5 /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,digvijay
digvijay@digvijay-virtualbox:~/workspace$
```

20. Search for the string "nologin" inside the `/etc/passwd` file and count how many lines contain it.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ grep "nologin" /etc/passwd
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,,:/:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
tcpdump:x:109:112::/nonexistent:/usr/sbin/nologin
cups-pk-helper:x:111:113:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:990:990:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:113:116::/var/lib/saned:/usr/sbin/nologin
geoclue:x:114:117::/var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:115:113::/nonexistent:/usr/sbin/nologin
polkitd:x:989:989:User for polkitd:/:/usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
colord:x:118:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
nm-openvpn:x:119:121:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
digvijay@digvijay-virtualbox:~/workspace$
```

21. Find all files in the `/var` directory that end with the extension `.log`.

```
digvijay@digvijay-virtualbox: /var
digvijay@digvijay-virtualbox:/$ ls
bin                boot  etc   lib    lib.usr-is-merged  media  opt   root  sbin               snap  swapfile  tmp  var
bin.usr-is-merged  dev   home  lib64  lost+found         mnt    proc  run   sbin.usr-is-merged  srv   sys       usr
digvijay@digvijay-virtualbox:/$ cd var/
digvijay@digvijay-virtualbox:/var$ sudo find /var -name "*.log"
[sudo] password for digvijay:
/var/log/boot.log
/var/log/gpu-manager.log
/var/log/installer/casper.log
/var/log/apt/history.log
/var/log/apt/term.log
/var/log/unattended-upgrades/unattended-upgrades-shutdown.log
/var/log/wvdialconf.log
/var/log/dpkg.log
/var/log/alternatives.log
/var/log/bootstrap.log
/var/log/apport.log
/var/log/Xorg.0.log
/var/log/auth.log
/var/log/kern.log
/var/log/sddm.log
/var/log/fontconfig.log
digvijay@digvijay-virtualbox:/var$
```

22. Create a symbolic link in your home directory named `link_to_delta` that points to `workspace/backup/delta.txt`.

```
digvijay@digvijay-virtualbox: ~/workspace/backup
digvijay@digvijay-virtualbox:~/workspace/backup$ ln -s ~/workspace/backup/delta.txt ~/list_to_delta
digvijay@digvijay-virtualbox:~/workspace/backup$ ls -l ~/list_to_delta
lrwxrwxrwx 1 digvijay digvijay 41 Feb 27 23:39 /home/digvijay/list_to_delta -> /home/digvijay/workspace/backup/delta.txt
digvijay@digvijay-virtualbox:~/workspace/backup$
```

23. Create a hidden file named `.secret` in your home directory.

```
digvijay@digvijay-virtualbox: ~
digvijay@digvijay-virtualbox:~$ touch .secret
digvijay@digvijay-virtualbox:~$ ls -a
.                bda         list_to_delta  .secret                                  .vboxclient-display-svga-x11-tty2-control.pid  .vboxclient-seamless-tty2-service.pid
..               .cache      .local         snap                                     .vboxclient-display-svga-x11-tty2-service.pid  .vboxclient-vmsvga-session-tty2-control.pid
Assignment       .config     Music          .sudo_as_admin_successful                .vboxclient-draganddrop-tty2-control.pid       Videos
.bash_history    Desktop     Pictures       Templates                                .vboxclient-draganddrop-tty2-service.pid       workspace
.bash_logout     Documents   .profile       .vboxclient-clipboard-tty2-control.pid   .vboxclient-hostversion-tty2-control.pid       .xsession-errors
.bashrc          Downloads   Public         .vboxclient-clipboard-tty2-service.pid   .vboxclient-seamless-tty2-control.pid
digvijay@digvijay-virtualbox:~$
```

24. List all files in your home directory, including hidden files, in a long format ($ls -la$).

```
digvijay@digvijay-virtualbox: ~
digvijay@digvijay-virtualbox:~$ ls -la
total 1012
drwxr-x--- 17 digvijay digvijay   4096 Feb 27 23:40 .
drwxr-xr-x  4 root     root       4096 Feb 27 22:43 ..
drwxrwxr-x  3 digvijay digvijay   4096 Feb 27 21:03 Assignment
-rw-------  1 digvijay digvijay   1224 Feb 27 22:22 .bash_history
-rw-r--r--  1 digvijay digvijay    220 Mar 31  2024 .bash_logout
-rw-r--r--  1 digvijay digvijay   3771 Mar 31  2024 .bashrc
drwxrwxr-x  2 digvijay digvijay   4096 Feb 26 23:35 bda
drwxrwxr-x  9 digvijay digvijay   4096 Feb 27 21:13 .cache
drwxrwxr-x 10 digvijay digvijay   4096 Feb 26 23:35 .config
drwxrwxr-x  2 digvijay digvijay   4096 Feb 27 20:58 Desktop
drwxr-xr-x  2 digvijay digvijay   4096 Feb 26 23:26 Documents
drwxr-xr-x  2 digvijay digvijay   4096 Feb 26 23:26 Downloads
lrwxrwxrwx  1 digvijay digvijay     41 Feb 27 23:39 list_to_delta -> /home/digvijay/workspace/backup/delta.txt
drwxrwxr-x  4 digvijay digvijay   4096 Feb 26 23:26 .local
drwxr-xr-x  2 digvijay digvijay   4096 Feb 26 23:26 Music
drwxr-xr-x  2 digvijay digvijay   4096 Feb 26 23:26 Pictures
-rw-r--r--  1 digvijay digvijay    807 Mar 31  2024 .profile
drwxr-xr-x  2 digvijay digvijay   4096 Feb 26 23:26 Public
-rw-rw-r--  1 digvijay digvijay      0 Feb 27 23:40 .secret
drwx------  4 digvijay digvijay   4096 Feb 27 22:16 snap
-rw-r--r--  1 digvijay digvijay      0 Feb 26 23:27 .sudo_as_admin_successful
drwxr-xr-x  2 digvijay digvijay   4096 Feb 26 23:26 Templates
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-clipboard-tty2-control.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-clipboard-tty2-service.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-display-svga-x11-tty2-control.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-display-svga-x11-tty2-service.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-draganddrop-tty2-control.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-draganddrop-tty2-service.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-hostversion-tty2-control.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-seamless-tty2-control.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-seamless-tty2-service.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-vmsvga-session-tty2-control.pid
drwxr-xr-x  2 digvijay digvijay   4096 Feb 26 23:26 Videos
drwxrwxr-x  3 digvijay training   4096 Feb 27 23:16 workspace
-rw-------  1 digvijay digvijay 904400 Feb 27 23:41 .xsession-errors
digvijay@digvijay-virtualbox:~$
```

25. Display the current system date and time and append it to a file named `timestamp.txt`.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ ls
backup  etc_list.txt
digvijay@digvijay-virtualbox:~/workspace$ date > timestamp.txt
digvijay@digvijay-virtualbox:~/workspace$ ls
backup  etc_list.txt  timestamp.txt
digvijay@digvijay-virtualbox:~/workspace$ cat timestamp.txt
Friday 27 February 2026 11:44:47 PM IST
digvijay@digvijay-virtualbox:~/workspace$
```

26. Display the name of the current logged-in user.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ who
digvijay tty2         2026-02-27 20:57 (:0)
digvijay@digvijay-virtualbox:~/workspace$ whoami
digvijay
digvijay@digvijay-virtualbox:~/workspace$ logname
digvijay
digvijay@digvijay-virtualbox:~/workspace$ echo $USER
digvijay
digvijay@digvijay-virtualbox:~/workspace$
```

27. Display the path of your current working directory.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ pwd
/home/digvijay/workspace
digvijay@digvijay-virtualbox:~/workspace$
```

28. Use the `df` command to check disk space in human-readable format and save it to `disk_usage.txt`.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ df
Filesystem     1K-blocks    Used Available Use% Mounted on
tmpfs             813324    1312    812012   1% /run
/dev/sda1       25621472 7786756  16507884  33% /
tmpfs           4066604       0   4066604   0% /dev/shm
tmpfs              5120       8      5112   1% /run/lock
tmpfs            813320     120    813200   1% /run/user/1000
digvijay@digvijay-virtualbox:~/workspace$ df > disk_usage.txt
digvijay@digvijay-virtualbox:~/workspace$ cat disk_usage.txt
Filesystem     1K-blocks    Used Available Use% Mounted on
tmpfs             813324    1312    812012   1% /run
/dev/sda1       25621472 7786760  16507880  33% /
tmpfs           4066604       0   4066604   0% /dev/shm
tmpfs              5120       8      5112   1% /run/lock
tmpfs            813320     120    813200   1% /run/user/1000
digvijay@digvijay-virtualbox:~/workspace$
```

29. Use the `du` command to check the size of the `workspace` directory.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ du ~/workspace/
4       /home/digvijay/workspace/backup
32      /home/digvijay/workspace/
digvijay@digvijay-virtualbox:~/workspace$
```

30. Identify your system's IP address using the command line.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fd17:625c:f037:2:e345:8901:bfe4:7dae  prefixlen 64  scopeid 0x0<global>
        inet6 fe80::a2b0:c59f:9c53:8b9c  prefixlen 64  scopeid 0x20<link>
        inet6 fd17:625c:f037:2:c843:b384:d250:c8fb  prefixlen 64  scopeid 0x0<global>
        ether 08:00:27:1b:ce:ff  txqueuelen 1000  (Ethernet)
        RX packets 124932  bytes 178741734 (178.7 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 10961  bytes 935081 (935.0 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 578  bytes 66409 (66.4 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 578  bytes 66409 (66.4 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

digvijay@digvijay-virtualbox:~/workspace$
```